Introduction to Artificial intelligence
Tutorial Sheet 1 (correction)

1.1 Define in your own words: (a) intelligence, (b) artificial intelligence, (c) agent, (d) rationality, (e) logical reasoning.

a. Dictionary definitions of intelligence talk about "the capacity to acquire and apply knowledge" or "the faculty of thought and reason" or "the ability to comprehend and profit from experience." These are all reasonable answers, but if we want something
quantifiable we would use something like "the ability to act successfully across a wide range of objectives in complex environments." "the ability to solve complex problems or make decisions with outcomes benefiting the actor"
b. We define artificial intelligence as the study and construction of agent programs that perform well in a given class of environments, for a given agent architecture; they do the right thing. An important part of that is dealing with the uncertainty of what the current state is, what the outcome of possible actions might be, and what is it that we really desire. "the ability of a digital computer or program to perform tasks commonly associated with intelligent beings"
c. We define an agent as an entity that takes action in response to percepts from an environment.
d. We define rationality as the property of a system which does the "right thing" given what it knows. See Section 2.2 for a mor complete discussion. The basic concept is perfect rationality; Section ?? describes the impossibility of achieving perfect rationality and proposes an alternative definition. **"a decision-making process that is based on making choices that result in the optimal level of benefit or utility for an individual"**
e. We define logical reasoning as the a process of deriving new sentences from old, such that the new sentences are necessarily true if the old ones are true. (Notice that does not refer to any specific syntax or formal language, but it does require a well-defined notion of truth.) "involves using a given set of facts or data to deduce other facts by reasoning logically"

1.2 Are reflex actions (such as flinching from a hot stove) rational? Are they intelligent?

Yes, reflex action are rational, in this case a slower, deliberative actions would tend to result in more damage to the hand. If "intelligent" means "applying knowledge" or "using thought and reasoning" then it does not require intelligence to make a reflex action

1.3 The Turing Test[1] is a deceptively simple method of determining whether a machine can demonstrate human intelligence: If a machine can engage in a conversation with a human without being detected as a machine, it has demonstrated human intelligence. The test is conducted in an interrogation room run by a judge. The test subjects, a person and a computer program, are hidden from view. The judge has a conversation with both parties and attempts to identify which is the human and which is the computer, based on the quality of their conversation. Turing concludes that if the judge can't tell the difference, the computer has succeeded in demonstrating human intelligence. That is, it can think. He predicts that, by the year 2000, a computer will have a 30% chance of passing a five-minute Turing Test with an unskilled interrogator. What chance do you think a computer would have today?

It's hard to give a definitive "chance" of a computer passing the Turing Test. While modern AI could likely deceive some judges in specific testing conditions, especially with limited interactions, the conversation's depth and the judge's expertise would play crucial roles. The test itself also has limitations, raising questions about whether it's the best measure of true intelligence.

Limitations of the Turing Test:

- Mimicry vs. Understanding: Passing the Turing test often involves mimicking human responses, making it more a test of mimicry and language generation than genuine human-like reasoning. Many AI systems excel at mimicking human conversations but lack deep reasoning capabilities.

[1] Read Turing's original paper on AI (Turing, 1950): https://academic.oup.com/mind/article/LIX/236/433/986238

- Lack of Self-Awareness: The Turing test does not require AI to be self-aware or have an understanding of its own reasoning. It focuses solely on external interactions and responses, neglecting the introspective aspect of human cognition.
- Failure to Address Thinking: Alan Turing himself recognized that the test might not truly address the question of whether machines can think. The test is more about imitation than cognition.

1.4 The neural structure of the sea slug *Aplysia* has been widely studied (first by Nobel Laureate Eric Kandel) because it has only about 20,000 neurons, most of them large and easily manipulated. Assuming that the cycle time for an *Aplysia* neuron is roughly the same as for a human neuron, how does the computational power, in terms of memory updates per
second, compare with the high-end computer?

	Supercomputer	Personal Computer	Human Brain
Computational units	10^4 CPUs, 10^{12} transistors	4 CPUs, 10^9 transistors	10^{11} neurons
Storage units	10^{14} bits RAM	10^{11} bits RAM	10^{11} neurons
	10^{15} bits disk	10^{13} bits disk	10^{14} synapses
Cycle time	10^{-9} sec	10^{-9} sec	10^{-3} sec
Operations/sec	10^{15}	10^{10}	10^{17}
Memory updates/sec	10^{14}	10^{10}	10^{14}

Answer:

The answer to this question is as follows:

The sea slug Aplysia has approximately the same cycle period as a human brain, namely 10^{-3}.

So the computational power in terms of memory updates per second is: $\frac{20{,}000}{10^{-3}} = 2 \times 10^7$ memory updates per second.

The high-end computer computational power is 10^{14} memory updates per second, so a high-end computer is approximately twice as fast as the sea slug Aplysia.

1.5 To what extent are the following computer systems instances of artificial intelligence:

• Supermarket bar code scanners.
Although bar code scanning is in a sense computer vision, these are not AI systems. The problem of reading a bar code is an extremely limited and artificial form of visual interpretation, and it has been carefully designed to be as simple as possible, given the hardware.

• Web search engines.
In many respects. The problem of determining the relevance of a web page to a query is a problem in natural language understanding, and the techniques are related to those we will discuss in Chapters 23 and 24. Search engines also use clustering techniques analogous to those we discuss in Chapter 20. Likewise, other functionalities provided by search engines use intelligent techniques; for instance, the spelling corrector uses a form of data mining based on observing users' corrections of their spelling errors. On the other hand, the problem of indexing billions of web pages in a way that allows retrieval in seconds is a problem in database design, not in artificial intelligence.

• Voice-activated telephone menus.
To a limited extent. Such menus tend to use very limited vocabularies – e.g. the digits, "Yes", and "No" — and within the designers' control, which greatly simplifies the problem. On the other hand, the programs must deal with an uncontrolled space of all kinds of voices and accents. Modern digital assistants like Siri and the Google Assistant makes more use of artificial intelligence techniques but still has a limited repertoire.

• Spelling and grammar correction features in word processing programs.

Slightly at most. The spelling correction feature here is done by string comparison to a fixed dictionary. The grammar correction is more sophisticated as it needs to use a set of rather complex rules reflecting the structure of natural language, but still, this is a very limited and fixed task. The spelling correctors in search engines would be considered much more nearly instances of AI than the Word spelling corrector are, first, because the task is much more dynamic – search engine spelling correctors deal very effectively with proper names, which are detected dynamically from user queries – and, second, because of the technique used – data mining from user queries vs. string matching.

• Internet routing algorithms that respond dynamically to the state of the network.
This is borderline. There is something to be said for viewing these as intelligent agents working in cyberspace. The task is sophisticated, the information available is partial, the techniques are heuristic (not guaranteed optimal), and the state of the world is dynamic. All of these are characteristics of intelligent activities. On the other hand, the task is very far from those normally carried out in human cognition. In recent years, there have been suggestions to base more core algorithmic work on machine learning.

1.6 Is AI a science, or is it engineering? Or neither or both? Explain.
This question is intended to be about the essential nature of the AI problem and what is required to solve it, but could also be interpreted as a sociological question about the current practice of AI research. A science is a field of study that leads to the acquisition of empirical knowledge by the scientific method, which involves falsifiable hypotheses about what is. A pure engineering field can be thought of as taking a fixed base of empirical knowledge and using it to solve problems of interest to society. Of course, engineers do bits of science—e.g., they measure the properties of building materials—and scientists do bits of engineering to create new devices and so on. The "human" side of AI is clearly an empirical science—called cognitive science these days—because it involves psychological experiments designed out to find out how human cognition works. What about the "rational" side? If we view it as studying the abstract relationship among an arbitrary task environment, a computing device, and the program for that computing device that yields the best performance in the task environment, then the rational side of AI is really mathematics and engineering; it does not require any empirical knowledge about the actual world—and the actual task environment—that we inhabit; that a given program will do well in a given environment is a theorem. (The same is true of pure decision theory.) In practice, however, we are interested in task environments that approximate the actual world, so even the rational side of AI involves finding out what the actual world is like. For example, in studying rational agents that communicate, we are interested in task environments that contain humans, so we have to find out what human language is like. In studying perception, we tend to focus on sensors such as cameras that extract useful information from the actual world. (In a world without light, cameras wouldn't be much use.) Moreover, to design vision algorithms that are good at extracting information from the camera images, we need to understand the actual world that generates those images. Obtaining the required understanding of scene characteristics, object types, surface markings, and so on is a quite different kind of science from ordinary physics, chemistry, biology, and so on, but it is still science. In summary, AI is engineering, but it would not be especially useful to us if it were not also an empirical science concerned with those aspects of the real world that affect the design of intelligent systems for that world.

1.7 "Surely animals cannot be intelligent—they can do only what their genes tell them."
Is the latter statement true, and does it imply the former?
This depends on your definition of "intelligent" and "tell." In one sense computers only do what the programmers command them to do, but in another sense what the programmers consciously tells the computer to do often has very little to do with what the computer actually does. Anyone who has written a program with an ornery bug knows this, as does anyone who has written a successful machine learning program. So in one sense Samuel "told" the computer "learn to play checkers better than I do, and then play that way," but in another sense he told the computer "follow this learning algorithm" and it learned to play. So we're left in the situation where you may or may not consider learning to play checkers to be s sign of intelligence (or you may think that learning to play in the right way requires intelligence, but not in this way), and you may think the intelligence resides in the programmer or in the computer.

1.8 "Surely animals, humans, and computers cannot be intelligent—they can do only what their constituent atoms are told to do by the laws of physics." Is the latter statement true, and does it imply the former?

The point of this exercise is to notice the parallel with the previous one. Whatever you decided about whether computers could be intelligent in 1.11, you are committed to making the same conclusion about animals (including humans), *unless* your reasons for deciding whether something is intelligent take into account the mechanism (programming via genes versus programming via a human programmer). Note that Searle makes this appeal to mechanism in his Chinese Room argument (see Chapter 26).

1.9 There are well-known classes of problems that are intractably difficult for computers, and other classes that are provably undecidable. Does this mean that AI is impossible?

No. It means that AI systems should avoid trying to solve intractable problems. Usually, this means they can only approximate optimal behavior. Notice that humans don't solve NPcomplete problems either. Sometimes they are good at solving specific instances with a lot of structure, perhaps with the aid of background knowledge. AI systems should attempt to do the same.

1.10 Why would evolution tend to result in systems that act rationally? What goals are such systems designed to achieve?

Evolution tends to result in systems that act rationally because rational behavior often leads to better survival and reproduction outcomes. Organisms that can make rational decisions, such as assessing risks, finding resources efficiently, and avoiding danger, are more likely to thrive and pass on their genes to future generations. The goals of such systems designed by evolution are primarily aimed at enhancing the organism's chances of survival and reproduction within its environment. These goals include finding food, shelter, mates, avoiding predators, and successfully raising offspring. Essentially, the systems are designed to maximize the organism's fitness, which is its ability to survive and reproduce in its specific ecological niche.

1.11 Examine the AI literature to discover whether the following tasks can currently be solved by computers:

1. Playing a decent game of table tennis (Ping-Pong): Yes, there are AI systems that can play table tennis at a decent level. For example, there are AI-powered robots designed specifically for playing table tennis that can compete against human opponents.

2. Driving in the center of Algiers, Algeria: While self-driving technology has advanced significantly, driving in complex urban environments like Algiers remains challenging due to factors like chaotic traffic patterns, unpredictable pedestrian behavior, and lack of standardized road infrastructure. Progress is being made, but fully autonomous driving in such extreme environments is not yet widespread.

3. Driving in San Francisco, California: Yes, currently one of the few cities in the world where the public can hail an autonomous driverless robo-taxi.

4. Buying a week's worth of groceries at the market: No. No robot can currently put together the tasks of moving in a crowded environment, using vision to identify a wide variety of objects, and grasping the objects (including squishable vegetables) without damaging them. The component pieces are nearly able to handle the individual tasks, but it would take a major integration effort to put it all together.

5. Buying a week's worth of groceries on the Web: Yes, Software robots are capable of handling such tasks, particularly if the design of the web grocery shopping site does not change radically over time.

6. Discovering and proving new mathematical theorems: Yes. For example, the proof of Robbins algebra described on page 360. The computer proof of the Robbins Problem was completed on October 10, 1996 by a program named "EQP".

7. Writing an intentionally funny story: AI systems can generate text that can be humorous, but writing a genuinely funny story that resonates with human audiences in a nuanced way remains a challenging task. AI-generated humor often lacks the subtlety, context awareness, and understanding of cultural nuances that humans possess.

8. Translating spoken English into spoken French in real time: Real-time speech translation systems exist, and some can translate between English and French with decent accuracy. While they may not be perfect, they can facilitate basic communication in real-time scenarios.

9. Performing a complex surgical operation: Yes. Robots are increasingly being used for surgery, although always under the command of a doctor. Robotic skills demonstrated at superhuman levels include drilling holes in bone to insert artificial joints, suturing, and knot-tying. They are not yet capable of planning and carrying out a complex operation autonomously from start to finish.